VisionGuard: Presentation Script

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Who presents what

Speaker	Slides	Topics	Time
Darshan	1–11, 19, 25	Introduction, problem, objectives, block diagram, hardware, sensors, live demo, conclusion	~7:43
Tanay	12–18, 20–24, 25	Power, circuit (no breadboard), algorithm, DFG, state diagram, code, testing, cost, limitations, future scope	~6:27
Total			**~14:25**

Times are approximate. If you're short of time, shorten slides 8, 9, 14 and 24. Never skip the demo (slide 19).

#	Slide	Speaker	Time
1	Title	Both	0:25
2	What this presentation covers	Darshan	0:20
3	The problem with a white cane	Darshan	0:40
4	Existing solutions and their limits	Darshan	0:35
5	Objectives	Darshan	0:40
6	Proposed solution	Darshan	0:35
7	Block diagram	Darshan	0:40
8	Hardware components	Darshan	0:35
9	Why the Arduino Nano	Darshan	0:30
10	How the ultrasonic sensor measures distance	Darshan	0:45
11	How the tip detects water	Darshan	0:45
12	Power system	Tanay	0:35
13	Circuit diagram	Tanay	0:40
14	Pin connections	Tanay	0:20
15	Algorithm	Tanay	0:45
16	Data flow graph	Tanay	0:35
17	State diagram	Tanay	0:30
18	Key code logic	Tanay	0:40
19	Prototype and assembly + live demo	Darshan	1:15
20	Testing and results	Tanay	0:30
21	Cost of the cane hardware	Tanay	0:25
22	Advantages and limitations	Tanay	0:35
23	Future scope	Tanay	0:30
24	References	Tanay	0:10
25	Conclusion / Thank you	Both	0:25

Slide-by-slide script

Don't read word for word. Learn the idea of each slide and say it in your own words. Point at the screen or the cane while talking.

Slide 1: Title

Both · ~25 s

Action: Stand on either side of the screen. Darshan holds the cane.

DARSHAN Good morning, sir. I'm S Darshan, registration number 25BCE2883.

TANAY And I'm Tanay Agarwal, 25BCE2475.

DARSHAN Our project is VisionGuard, a smart cane for visually impaired people. It warns the user about obstacles in front of them and about water at the tip of the cane.

Slide 2: What this presentation covers

Darshan · ~20 s

DARSHAN First I'll explain the problem, our objectives, the system design and the sensors. Then Tanay will cover the power, circuit, software, results, cost and future scope.

Slide 3: The problem with a white cane

Darshan · ~40 s

Action: Point at the 2.2 billion figure, then the three points.

DARSHAN According to the WHO, about 2.2 billion people around the world have some kind of vision impairment. Most blind people use a white cane, but it has three problems. First, it finds an obstacle only when it actually hits it. Second, it can't warn about water, puddles or open drains. Third, it doesn't tell the user how far away the obstacle is.

Slide 4: Existing solutions and their limits

Darshan · ~35 s

Action: Move left to right across the three boxes.

DARSHAN We looked at what already exists. A white cane is cheap and simple, but it works only on contact. A guide dog is very helpful, but it's expensive, needs long training and isn't easily available. There are commercial smart canes, but they're costly, mostly imported and hard to repair here. So the gap we saw is a low-cost cane that warns early about both obstacles and water.

Slide 5: Objectives

Darshan · ~40 s

Action: Point at each number as you say it.

DARSHAN We set four objectives. One: detect obstacles within 50 centimetres. Two: show the distance through the beep speed. The gap between beeps goes from 50 milliseconds when the obstacle is very close to 400 milliseconds at 50 centimetres. Three: detect water when the moisture reading drops below 500 out of 1023, and give that the highest priority. Four: keep it cheap and portable. The cane parts cost about 876 rupees, and it runs on one battery.

Slide 6: Proposed solution

Darshan · ~35 s

Action: Hold up the cane next to the photo.

DARSHAN Our solution has three main parts on a PVC pipe: an ultrasonic sensor that works like the cane's eyes, a moisture probe at the tip, and a buzzer that gives alerts by priority. The idea actually started as a smart shoe. We moved it to a cane because the cane tip reaches water before the foot does, and a sensor on the shaft stays steadier than one on a moving shoe.

Slide 7: Block diagram

Darshan · ~40 s

Action: Trace the diagram left, middle, right, then top.

DARSHAN This is our block diagram. On the left are the inputs: the ultrasonic sensor connected to pins D5 and D6, and the moisture probe connected through the LM393 module to pin A0. In the middle is the Arduino Nano, which makes all the decisions. On the right is the output, the buzzer on pin D2. At the top, the battery powers everything through the on/off switch. The dashed box is a vibration motor, which we plan to add later.

Slide 8: Hardware components

Darshan · ~35 s

Action: Go across the top row, then the bottom row.

DARSHAN These are the eight parts we used. The Arduino Nano is the controller. The HC-SR04 measures distance. The moisture probe and the LM393 module detect water. The active buzzer gives the alerts. The 18650 battery, the rocker switch and a 220 microfarad capacitor make up the power side. The capacitor smooths the supply so the Nano doesn't reset when the buzzer switches on.

Slide 9: Why the Arduino Nano

Darshan · ~30 s

DARSHAN We chose the Arduino Nano because it's small, only 18 by 45 millimetres, so it fits on a cane. It costs about 250 rupees. It uses 5-volt logic like our sensors, and it has a built-in 10-bit ADC to read the moisture sensor. It's also easy to program over USB. A Raspberry Pi would be too expensive and power-hungry for a simple job like this.

Slide 10: How the ultrasonic sensor measures distance

Darshan · ~45 s

Action: Point at steps 1, 2, 3, then the formula box.

DARSHAN This is how the ultrasonic sensor works. Step one: the Nano sends a 10-microsecond pulse to the Trig pin. Step two: the sensor sends out eight sound pulses at 40 kilohertz, which is too high for us to hear. Step three: the Echo pin stays HIGH until the echo comes back. So distance equals speed of sound times time, divided by two. We divide by two because the sound goes to the object and comes back. The speed of sound is about 0.034 centimetres per microsecond, so if the echo takes 1000 microseconds, the distance is 17 centimetres.

Slide 11: How the tip detects water

Darshan · ~45 s

Action: Point at the probe on the real cane. End by turning towards Tanay.

DARSHAN This is the moisture probe at the tip. It has two copper strips. Dry air doesn't conduct electricity, but water does, so when the tip touches water, the resistance between the strips drops. The LM393 module turns this into a voltage, and the Nano reads it on A0. When it's dry the reading is about 1023, and when it's wet it goes below 500. We use the analog output instead of the digital D0 pin, so we can set the threshold in the code.

DARSHAN Now Tanay will explain the power system and the circuit.

Slide 12: Power system

Tanay · ~35 s

Action: Point at battery → switch → 5V pin, then the warning.

TANAY Thank you. The cane runs on one 18650 lithium-ion battery: 3.7 volts, 1200 milliamp-hours. It goes through the rocker switch to the Nano's 5V pin, which also powers both sensors. We didn't use the VIN pin, because VIN needs about 7 volts. The whole cane draws around 40 milliamps, so in theory the battery can last about 30 hours. One safety point: we always switch the cane off before connecting USB, because the battery is connected directly to the 5V line.

Slide 13: Circuit diagram

Tanay · ~40 s

Action: Point at the red 5V line and black GND line first.

TANAY This is the full circuit. The red line is 5 volts and the black line is ground. Every module shares them. Trig goes to D5, Echo to D6, the moisture signal to A0 and the buzzer to D2. In the final cane we didn't use a breadboard. We soldered male header pins onto the Nano, and since the sensors also have header pins, the signal wires are female jumper wires plugged onto both ends. The power wires, the buzzer and the 220 microfarad capacitor are soldered for a strong connection.

Slide 14: Pin connections

Tanay · ~20 s

TANAY This table lists every connection. The main pins to remember are D5 for Trig, D6 for Echo, A0 for moisture and D2 for the buzzer. The LM393's D0 pin isn't used.

Slide 15: Algorithm

Tanay · ~45 s

Action: Point at boxes 1 to 6 in order.

TANAY This is our algorithm. First the Nano sets up the pins. Then, in a loop, it measures the distance and reads the moisture. Then it decides. If the moisture is below 500, that's water, so the buzzer stays on continuously. If not, and an obstacle is closer than 50 centimetres, it beeps for 50 milliseconds and then waits a gap between 50 and 400 milliseconds depending on the distance. Otherwise the path is clear and the buzzer stays off. Then it waits 30 milliseconds and repeats.

Slide 16: Data flow graph

Tanay · ~35 s

Action: Trace the top path, then the bottom path, then the priority box.

TANAY The data flow graph shows how the data is processed. The echo time t is multiplied by 0.034 and divided by 2 to get the distance d. The distance is used in two places: to check whether the obstacle is near, and to work out the beep gap g. The moisture value m is compared with 500 to decide whether it's wet. Finally, the priority block uses wet, near and g to control the buzzer.

Slide 17: State diagram

Tanay · ~30 s

Action: Follow the arrows from OFF to SENSE, then the three states.

TANAY This is the state diagram. When the switch is turned on, the cane sets up once and goes to SENSE. From SENSE it goes into exactly one state: water alert, obstacle alert or clear. After 30 milliseconds it comes back to SENSE. Turning the switch off takes it back to OFF from any state.

Slide 18: Key code logic

Tanay · ~40 s

Action: Point at the top half of the code, then the bottom half. End by turning towards Darshan.

TANAY This is the main part of our code. The readDistanceCm function sends the trigger pulse and uses pulseIn with a 30-millisecond timeout. Without the timeout, it would wait up to one second when there's no echo and freeze the cane. Below that is the if-else chain. Water is checked first, so it always has priority. The map function converts the distance into the beep gap.

TANAY Now Darshan will show you the prototype.

Slide 19: Prototype and assembly + live demo

Darshan · ~75 s

Action: Point at the photos, then at the real cane. DEMO: switch on → hand from far to near → dip tip in water.

DARSHAN This is our prototype and how we put it together. We soldered male header pins onto the Nano and connected the sensors with female jumper wires: Trig to D5, Echo to D6 and the moisture output to A0. The power wires, the buzzer and the capacitor are soldered. All the parts are hot-glued to the PVC pipe, the ultrasonic sensor sits on a foam spacer, and the battery holder is cable-tied. From the bottom: the probe, the LM393 module, the ultrasonic sensor, the Nano with the buzzer, then the switch and battery near the handle.

DARSHAN Let me show you. I'm switching it on. It's silent now because the path is clear. Now I'll move my hand closer. It starts beeping at 50 centimetres and gets faster as I get closer. Now I'll dip the tip in water. It gives a continuous beep, which means water is detected.

DARSHAN Tanay will now show our test results.

Slide 20: Testing and results

Tanay · ~30 s

Action: Point at the Result column.

TANAY These are our tests. Running on battery and the obstacle alert are both verified. The water alert is in the code, and we're calibrating the exact threshold using the dry and wet readings on the Serial Monitor. We'll update this table once that's done.

Slide 21: Cost of the cane hardware — Tanay · ~25 s

Action: Point at ₹876.

TANAY The parts on the cane cost about 876 rupees. Our total spending was 1,805 rupees, but that includes tools like the soldering iron and extra parts for the next version, like the MOSFET and the fuse. That's much cheaper than commercial smart canes.

Slide 22: Advantages and limitations — Tanay · ~35 s

Action: Left box first, then right box.

TANAY The advantages: it warns before the user hits an obstacle, it shows the distance through the beep speed, it detects water, and it's cheap and easy to repair. The limitations, to be honest: the sensor's beam is narrow, so it can miss head-level objects. Water is detected only when the tip touches it. The probe will corrode over time. The battery isn't regulated. And the buzzer can be hard to hear in traffic.

Slide 23: Future scope — Tanay · ~30 s

TANAY To fix these, we plan to add a vibration motor using the MOSFET, so the user can feel the alerts. We also plan proper power with a boost converter, a charger and a fuse; extra sensors pointing up and down for head-level objects and steps; a capacitive moisture sensor that won't corrode; GPS with an SOS message; and finally a PCB inside a waterproof box.

Slide 24: References — Tanay · ~10 s

TANAY These are the datasheets and documents we referred to.

Slide 25: Conclusion / Thank you — Both · ~25 s

Action: Both face the faculty.

DARSHAN To conclude, VisionGuard turns a normal cane into a low-cost electronic aid. It warns about obstacles early, shows the distance through the beep speed, and gives a priority warning for water, all for under 900 rupees.

TANAY Thank you, sir. We're happy to answer any questions.

Questions: who answers what

Topic of the question	Main answer	Backup
Problem, idea, why a cane instead of a shoe, objectives	Darshan	Tanay
Ultrasonic sensor, formula, 40 kHz, range, ÷ 2	Darshan	Tanay
Moisture probe, LM393, ADC, 1023 / 500	Darshan	Tanay
Arduino Nano, why not Uno or Raspberry Pi	Darshan	Tanay
Battery, 5V vs VIN, capacitor, safety	Tanay	Darshan
Wiring, soldering, no breadboard	Tanay	Darshan
Code: pulseIn, timeout, map(), priority, setup/loop	Tanay	Darshan
Testing, cost, limitations, future scope	Tanay	Darshan

If sir asks one of you a question on the other's topic, try to answer first. Then the other can add: *"Sir, to add to that..."*

Handover lines (practise these)

- Slide 11 → 12, Darshan: *"Now Tanay will explain the power system and the circuit."*
- Slide 18 → 19, Tanay: *"Now Darshan will show you the prototype."*
- Slide 19 → 20, Darshan: *"Tanay will now show our test results."*
- Slide 25, Darshan concludes, then Tanay: *"Thank you, sir. We're happy to answer any questions."*

Before you go in

- Run through the full script together **once out loud**, with a timer.
- Charge the battery. Carry the cane, the USB cable, a cup of water and a tissue.
- Test the demo once before entering: switch on, hand, water.
- If the battery dies: switch OFF, connect USB to the laptop, and demo from USB power.
- Remember: **d = t × 0.034 ÷ 2 · 50 cm · 500 · D5, D6, A0, D2**.
- If you don't know an answer: *"I'm not sure about that, sir. We'll check and get back to you."*